Supporting Information: Supporting Information: From African-Natural-Product Chemical Space to Resistance-Resilience Hypotheses

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This supplement contains the numerical material needed to reproduce the integrated computational analysis. It is organized by evidence layer: chemical space, target-anchored docking, RRS, and exploratory cross-metric analysis. The four-target matrix is reported target by target; no cross-target mean is used because the Vina score is not calibrated across unlike receptor pockets.

S1. Cohort and evidence boundaries

The integrated analysis uses the 17-member P2 Set-C polypharmacology-oriented cohort. The docking calculations follow AutoDock Vina conventions.[S1,S2] The MPO top-20 and the parent-study MD top-20 are distinct sets. The parent-study MD systems 201, 438, the 164-labelled PfClpR cohort, and 214 are not MD validation of Set C. The target panel contains four wild-type docking profiles, whereas RRS is calculated from the separate 'docking_mutants.csv‘ WT/mutant panel and is available only for PfDHFR and PfCRT mutant panels.

S2. Target-anchored docking matrix

Table S1: Target-anchored Vina estimates for the locked 17-member cohort. Values are in kcal mol^{-1}; they are not experimental free energies.

Candidate	PfDHFR	PfCRT	PfClpP	PfATP4
PP-01	−5.947	−6.483	−5.467	−6.361
PP-02	−5.562	−6.124	−5.907	−6.591
PP-03	−6.089	−6.633	−6.358	−7.311
PP-04	−5.493	−5.329	−5.386	−6.086
PP-05	−6.285	−6.412	−6.466	−6.378
PP-06	−6.267	−7.907	−7.031	−7.285
PP-07	−6.226	−6.633	−6.226	−5.775
PP-08	−4.935	−5.213	−5.052	−5.494
PP-09	−5.693	−6.871	−5.613	−6.078
PP-10	−6.346	−6.395	−6.016	−6.670
PP-11	−6.179	−6.376	−6.583	−6.724
PP-12	−5.248	−6.357	−5.386	−4.635
PP-13	−5.921	−5.685	−6.454	−7.565
PP-14	−4.863	−5.328	−5.047	−5.460
PP-15	−6.045	−6.084	−6.366	−7.016
PP-16	−5.469	−6.202	−6.203	−6.451
PP-17	−5.266	−5.116	−5.342	−5.360

S3. RRS definition and class summary

For candidate i, mutation m, and target t:

$$\mathrm{RRS}_{i,m,t} = 100|S_{\mathrm{Vina},i,m,t}|/|S_{\mathrm{Vina},i,WT,t}|.$$

Here S_{Vina} denotes an AutoDock Vina scoring-function output, not an experimental free energy. RRS eligibility is determined from the separate ‘docking_mutants.csv‘ WT/mutant panel, not from the four-target V5 wild-type matrix. Only genuine wild-type binders with

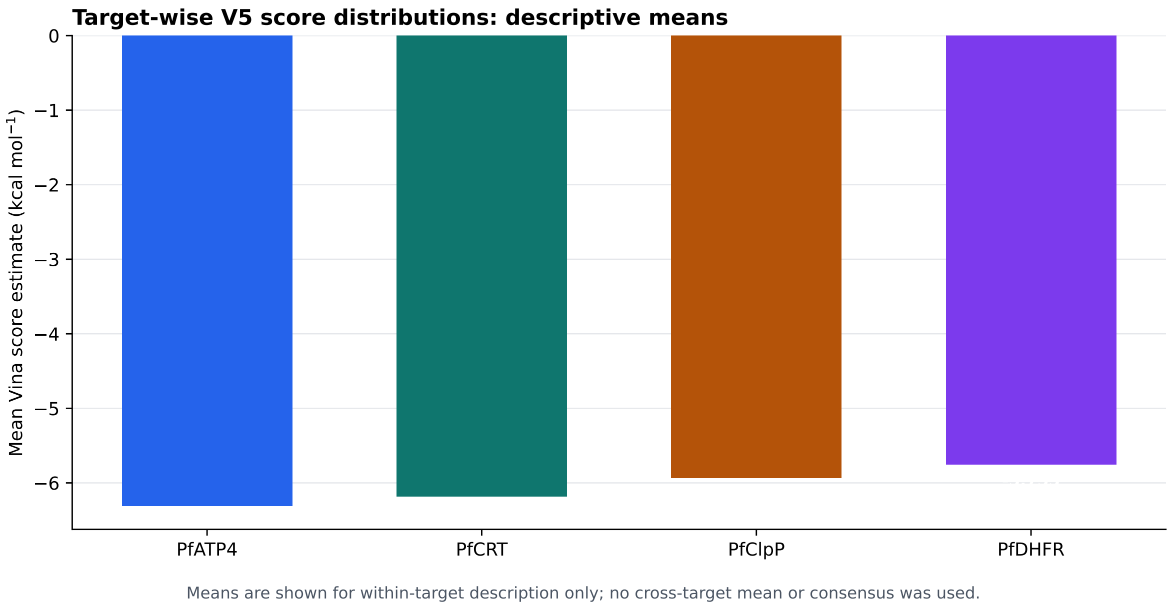

Figure S1: Descriptive target-wise means from the raw docking panel. Means are shown only within target; no cross-target arithmetic mean or consensus score was used. These summaries are computational scoring-function outputs, not experimental affinity measurements.

$|S_{\mathrm{Vina},i,WT,t}| \geq 5\,\mathrm{kcal\,mol^{-1}}$ contributed. The 17-candidate class counts were A*: 6, B: 5, C: 5, and D: 1. The observed RRS range was 68.2 to 111.7. The complete machine-readable values are in `Project2_Polypharmacology_MD_ValidationV2607/results/c_rrs_classification.csv`.

Table S2: RRS class definitions used in the integrated analysis.

Class	Operational definition	Numbe
A*	maximum absolute eligible WT score $\geq 7\,\mathrm{kcal\,mol^{-1}}$ and all available RRS values $\geq 80\,\%$	6
A	All available RRS values $\geq 80\,\%$, without the A* wild-type criterion	0
B	All available RRS values $\geq 70\,\%$, but not all $\geq 80\,\%$	5
C	At least one RRS value $\geq 80\,\%$, with the complete-panel criteria for A/A* or B not met	5
D	No available RRS value $\geq 80\,\%$	1

Table S3: Spearman correlations on the 17-member cohort. Bonferroni-adjusted threshold: $\alpha = 0.017$ for the three hypothesis-oriented comparisons.

Pair	ρ	p	Interpretation
PNS–RRS	-0.559	0.020	Nominal trend; not Bonferroni-significant
ACSI–RRS	-0.132	0.613	Not significant
RRS–wild-type Vina score	-0.433	0.082	Not significant
PNS–wild-type Vina score	0.389	0.123	Descriptive/mechanical comparison

S4. Exploratory cross-metric analysis

S5. Target evidence classes and limitations

PfDHFR uses a catalytic-site ligand anchor; PfClpP uses a catalytic triad; PfCRT uses a membrane-associated Y01 cavity proxy; and PfATP4 uses conserved P-type ATPase machinery without a co-crystallized inhibitor. Consequently, all four-target profiles are exploratory and are not equivalent evidence of biological target engagement. No RRS values are reported for PfClpP or PfATP4.

S6. Reproducibility and availability

The integrated candidate manifest, raw docking table, P2 RRS table, cross-metric matrix, scripts, receptor files, ligand files, and provenance records are available in the repository at https://github.com/NanaEngo/Malaria_codesV2. The figures and derived tables in this supplement are reproducible computational outputs; automated integrity checks document their derivation but do not substitute for independent structural or experimental validation.

References

(S1) Trott, O.; Olson, A. J. AutoDock Vina: Improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *Journal of*

Computational Chemistry **2009**, *31*, 455–461.

(S2) Eberhardt, J.; Santos-Martins, D.; Tillack, A. F.; Forli, S. AutoDock Vina 1.2.0: New Docking Methods, Expanded Force Field, and Python Bindings. *Journal of Chemical Information and Modeling* **2021**, *61*, 3891–3898.